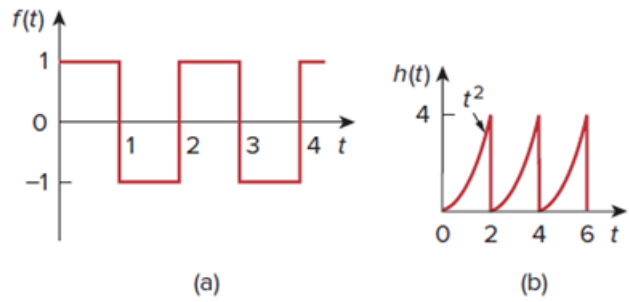


Problem

Determine the Laplace transforms of the periodic functions in Fig. 15.35.

Figure 15.35:



Step-by-step solution

Step 1 of 4

(a)

Refer to Figure 15.35 (a) in the textbook.

The period of the function is $T = 2$.

The expression for the signal $f(t)$ is,

$$f(t) = \begin{cases} 1 & \text{for } 0 < t < 1 \\ -1 & \text{for } 1 < t < 2 \end{cases}$$

Write the first period of the function in terms of step function.

$$\begin{aligned} f_1(t) &= [u(t) - u(t-1)] - [u(t-1) - u(t-2)] \\ &= u(t) - u(t-1) - u(t-1) + u(t-2) \\ &= u(t) - 2u(t-1) + u(t-2) \end{aligned}$$

Step 2 of 4

Apply Laplace transforms and time-shift property.

$$\begin{aligned} F_1(s) &= \frac{1}{s} - \frac{2e^{-s}}{s} + \frac{e^{-2s}}{s} \\ &= \frac{1}{s} (1 - 2e^{-s} + e^{-2s}) \\ &= \frac{1}{s} (1 - e^{-s})^2 \end{aligned}$$

Calculate the Laplace transform of the periodic function.

$$F(s) = \frac{F_1(s)}{1 - e^{-Ts}}$$

Substitute $\frac{1}{s}(1 - e^{-s})^2$ for $F_1(s)$ and 2 for T .

$$F(s) = \frac{1}{s} \frac{(1 - e^{-s})^2}{(1 - e^{-2s})}$$

Therefore, the Laplace transform of the periodic signal is $\boxed{\frac{1}{s} \frac{(1 - e^{-s})^2}{(1 - e^{-2s})}}$.

Step 3 of 4

(b)

Refer to Figure 15.35 (b) in the textbook.

The period of the function is $T = 2$.

The expression for the signal $h(t)$ is,

$$h(t) = \begin{cases} t^2 & \text{for } 0 < t < 2 \end{cases}$$

Write the first period of the function in terms of step function.

$$\begin{aligned} h_1(t) &= t^2 [u(t) - u(t-2)] \\ &= t^2 u(t) - t^2 u(t-2) \\ &= t^2 u(t) - (t-2+2)^2 u(t-2) \\ &= t^2 u(t) - [(t-2)^2 + 4(t-2) + 4] u(t-2) \\ &= t^2 u(t) - (t-2)^2 u(t-2) - 4(t-2) u(t-2) - 4u(t-2) \end{aligned}$$

Step 4 of 4

Apply Laplace transforms and time-shift property.

$$\begin{aligned}
 H_1(s) &= \frac{2}{s^3} [1 - e^{-2s}] - \frac{4}{s^2} (e^{-2s}) - \frac{4}{s} (e^{-2s}) \\
 &= \frac{2}{s^3} [1 - e^{-2s}] - \frac{4}{s^2} (e^{-2s}) - \frac{4}{s} (e^{-2s}) \\
 &= \frac{1}{s^3} [2(1 - e^{-2s}) - 4se^{-2s} - 4s^2e^{-2s}] \\
 &= \frac{1}{s^3} [2(1 - e^{-2s}) - 4e^{-2s}(s + s^2)]
 \end{aligned}$$

Calculate the Laplace transform of the periodic function.

$$H(s) = \frac{H_1(s)}{1 - e^{-Ts}}$$

Substitute $\frac{1}{s^3} [2(1 - e^{-2s}) - 4e^{-2s}(s + s^2)]$ for $H_1(s)$ and 2 for T .

$$H(s) = \frac{1}{s^3} \frac{[2(1 - e^{-2s}) - 4e^{-2s}(s + s^2)]}{(1 - e^{-2s})}$$

Therefore, the Laplace transform of the periodic signal is

$$\boxed{\frac{1}{s^3} \frac{[2(1 - e^{-2s}) - 4e^{-2s}(s + s^2)]}{(1 - e^{-2s})}}.$$